Text Generator Lab

Problem and solution. A small model can generate fluent-looking text without understanding it. We compare a plain POS-HMM, a role-split POS-HMM, and a trigram. Models train on 960 synthetic sentences; 240 held-out sentences are split by document. The Streamlit test bench samples batches, shows tags, computes metrics, exports CSV, and tags input with log-space Viterbi. These metrics describe the tiny synthetic grammar, not real English quality.

Generation at $T = 1.0$. Each model generates 500 sentences with seed 42. A valid pattern is a coarse tag sequence seen in training; a plausible pair is a first noun–verb pair seen in training; novel means the exact sentence is absent from training. Distinct-2 is the fraction of unique adjacent word pairs.

Model	Valid pattern	Plausible pair	Novel	Distinct-2
Plain HMM	16.2%	14.8%	99.8%	0.198
Role-split HMM	100.0%	60.6%	99.4%	0.148
Trigram	57.0%	76.0%	91.2%	0.105

The plain HMM reuses POS states at different positions and can combine incompatible transitions. Role-split states preserve subject, object, or prepositional roles, yielding 100% for this sample and corpus-defined metric. Trigram tags are estimated with the role-split HMM.

Temperature. For a writing toy, choose $T = 1.5$ for variety; for autocomplete, choose $T = 0.5$ to favour common continuations. These are heuristics, not measured autocomplete results.

Examples. For the role-split HMM, seed 42 and $T = 1$, "the children eat the bread at the park ." is a plausible novel sentence. "the farmers eat the river by the school ." has a valid pattern but fails in meaning: tag transitions and independent emissions do not encode semantic compatibility. Its subject–verb pair was seen, illustrating the proxy's limits.

Real data. Both HMMs achieved 100% token accuracy on the 240 held-out synthetic sentences. The same plain HMM and Viterbi on Universal Dependencies English EWT (12,544 train / 2,077 test; CC BY-SA 4.0) produced:

Model	Accuracy	Known	Unknown	Whole sentence
Most-frequent tag	86.2%	90.8%	28.6%	31.9%
HMM + Viterbi	88.0%	92.9%	28.2%	40.4%

Unknown words account for 7.5% of test tokens. Web text has 17 tags, many more patterns, ambiguity, and unseen words; smoothed emissions lack lexical features for unknowns. Synthetic text has six tags and a restricted grammar. State splitting improves positional structure but does not teach meaning or guarantee real-world tagging. Treebank revisions can change results.